

The Naval Architecture of

Noahs' Ark:

An Engineering

Feasibility Study

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An Engineering Feasibility Study of Noah's Ark.

Objective: To determine if the biblical account of the ark describes a vessel capable of safely bringing Noah and his family and the animals through the great flood.

Ground Rules:

- 1. Take the biblical description of the Noah's Ark and the flood at face value**
- 2. Use current engineering knowledge of ships to interpret the biblical data and to rough out a design for the ark as a basis for design calculations**

Background:

The biblical account as we know it was written by Moses about 1500 B.C. (About 3500 years ago)

The flood occurred, according to the Biblical dating (Usher's) about 2500 B.C. (About 4500 years ago)

Noah's Ark as a Ship

- **Builder's specifications, (Genesis 6:14 –16)**
Principal dimensions
Building materials
Internal arrangements
- **Description of the crew and cargo (animals) and cargo loading, (Genesis 6:18 – 7:9)**
Crew/passengers (Eight people: Gen. 6:18)
Livestock
Provisions
- **The voyage during the flood and subsequent grounding, (Genesis 7:11 – 8:13)**
“Launching”
Voyage: Ship's log data-weather, navigation
Grounding, waiting for proper time to unload/disembark

- **The disembarkation or unloading of the ark,**
(Genesis 8:13 – 19)
Preparations
Unloading

Naval Architectural Design Considerations

- **Cargo-Carrying Capacity**
- **Arrangements**
- **Hull Design**
- **Structural Design**
- **Hydrostatic Stability**
- **Resistance and Propulsion**
- **Seakeeping**
- **Stability and Control**

Popular Misconceptions

It took 100 (or 120) years to build the ark.

Noah was ridiculed while building the ark

The ark was built near Mt. Ararat.

The Great Flood as a “Local Flood”

Builders Specifications

Gen. 6:14-15

God says to Noah:*“Make yourself an ark of gopherwood; make rooms in the ark, and cover it inside and outside with pitch.”*

Principal dimensions: *“The length of the ark shall be three hundred cubits, its width fifty cubits, and its height thirty cubits.”* Length (L), Beam (B), and Depth (D)

Ventilation: *“You shall make a window for the ark, and you shall finish it to a cubit from above;”*

Loading: *“...and set the door of the ark in its side.”*

Arrangements: *“You shall make it with lower, second, and third decks.”*

The Voyage of the Ark

From Noah's ship's log:

Start of Flood: Gen 7:11: *“In the six hundredth year of Noah's life, in the second month, the seventeenth day of the month, on that day all the fountains of the great deep were broken up, and the windows of heaven were opened.”*

Gen. 7:12 as: *“And the rain was on the earth forty days and forty nights.”*

The Voyage Afloat

Reference to the draft of the ark (ship):

“The waters prevailed fifteen cubits upward, and the mountains were covered” (Gen 7:20)

The ark *“moved about on the surface of the waters”*

Five month duration of the voyage afloat: Gen 7:24: *“And the waters prevailed on the earth one hundred and fifty days”.*

The Post-Grounding Period

Gen 8:4 *“Then the ark rested on the seventh month, the seventeenth day of the month, on the mountains of Ararat.”*

Five months and 13 days after grounding, Gen 8:13 says *“And it came to pass in the six hundred and first year, in the first month, the first day of the month, that the waters were dried up from the earth; and Noah removed the covering of the ark and looked, and indeed the surface of the ground was dry.”*

The ark rested on the ground 220 days, or seven months, ten days.

Disembarkation

The date of the start of the disembarkation was one month twenty seven days after the ark's covering was removed according to Gen 8:14-16: *“And in the second month, on the twenty-seventh day of the month, the earth was dried. Then God spoke to Noah, Saying, “Go out of the ark, you and your wife...”*

The total time from being sealed up at the start of the flood to the time of disembarkation was 370 days, or one year and ten days.

Proportions and Size of the Ark

- **Length (L) = 300 cubits**
- **Breadth (B) = 50 cubits**
- **Depth (D) = 30 cubits**

One Cubit = 18 inches (1.5 ft.)

- **Length (L) = 450 ft.**
- **Breadth (B) = 75 ft.**
- **Depth (D) = 45 ft.**

Draft, Displacement of the Ark

(Estimated)

One-half the depth (15 cubits) or 22.5 ft.

Block coefficient = 0.75

Ark's displacement in salt water = 16,300 long tons.

Draft to Depth Ratio (T/D)

Ark: $T/D = 0.50$,

Ship A: $T/D = 0.634$

Ship C: $T/D = 0.624$

Ratios or Hull Proportions

- **$L / B = 300 / 50 = 6.0$**
- **$L / D = 300 / 30 = 10.0$**
- **$B / D = 50 / 30 = 1.67$**

Comparison of the Ark Dimensions with Typical Modern Ships (Arnott, 1955)

Dimension	Typical Cargo Vessel (Smaller)*	Typical Cargo Passenger Vessel *	Noah's Ark
Length (L) ft.	435	465	450
Breadth (B) ft.	63	69.5	75
Depth (D) ft.	40.5	42.5	45

Comparison of the Ark's Proportions with Modern Ships

Dimension Ratio	Cargo Vessel	Cargo Passenger Vessel C	Noah's Ark
L / B	6.91	6.67	6.00
L / D	10.74	10.94	10.00
B / D	1.56	1.64	1.667

Cargo-Carrying Capacity

Volume Calculation

$$\text{Internal Volume} = 450 * 75 * 45 * 0.8 = 1,215,000 \text{ cu. ft.}$$

$$\text{Horse Stall Volume: } 8 \text{ ft} \times 10 \text{ ft} \times 8 \text{ ft} = 640 \text{ cu. ft.}$$

$$\text{Number of Horses} = 1,215,000 * 0.5 / 640 = 950$$

Combinations:- 200 horse-size animals and about 15,000 smaller to medium size animals.

General Arrangements

Six decks:

- 1.First structural deck (above stone ballast and storage spaces)**
- 2.Lower stall/accommodation (non-structural) deck**
- 3.Second structural deck**
- 4.Upper stall/accommodation (non-structural) deck**
- 5.Third structural deck (the eight people occupy one of the compartments on this deck)**
- 6.Main (structural) deck**

Structural Design

The largest all-wooden ship: “Down Easter” Henry B. Hyde, built in 1884. The H. B. Hyde was 267.8 ft x 45ft.x28.8 ft (Chapelle, 1935). The ark has about 4.4 times the enclosed volume of the H. B. Hyde.

A midship section drawing of a wooden sailing ship from Arnott, (1955)

Carvel construction introduced by the Portugese around 1400 AD.

Pre-flood iron age:

Cain's geneology, Gen 5:22: *“Tubal-Cain, an instructor of every craftsman in bronze and iron”.*

(Possibly)

Iron (and Bronze) Fastenings, Hardware, Bracing

Physical Properties of Gopher Wood

Resinous wood, possibly cypress: Use hard pine or southern pine

From Gere and Timoshenko (1995): southern pine:

Property	Value	Units
Weight density	35-40	Lbs/ft ³
Modulus of elasticity (E)	1600-2000	ksi
Bending yield stress	6-9	ksi
Bending ultimate stress	8-14	ksi

Took Yield Stress for the wood planking as 6 ksi: one-sixth the strength of 36 ksi structural steel.

Longitudinal Strength

Simplified Ark Midship Section

1983 ABS Rules for Steel Vessels.

ABS Longitudinal Strength calculations:

The Nominal Permissible Bending Stress:

$$f_p = 22.75 \text{ (ksi).}$$

Minimum Section Modulus:

$$SM_{\min} = 28,471 \text{ (in}^2 \text{ ft)}$$

Still-Water Bending Moment:

$$M_s = 12,516 \text{ (in}^2 \text{ ft)}$$

Wave-Induced Bending Moment Amidships:

$$M_w = 173,529 \text{ (in}^2 \text{ ft)}$$

Total Bending Moment:

$$M_t = 298,044 \text{ (in}^2 \text{ ft)}$$

Section Modulus (SM):

$$SM/t = 785,040 \text{ (ft}^3 \text{)}$$

$$s = M_t / (SM)$$

For 36 ksi Steel, Planking Thickness (t):
:

$$t = 0.449 \text{ (in)}$$

For 6 ksi Wood the Planking Thickness is:

$$t = 2.7 \text{ (in)}$$

Naval Architectural Considerations

Resistance, and Propulsion

There is no mention of any kind of propulsion system for the ark.

“the ark “moved about on the surface of the waters”.

“...God made a wind to pass over the earth, and the waters subsided.”

The ark drifted downwind and with the currents while it was afloat.

The crew, was sealed up inside the ark:“...*the Lord shut him in.*”

My interpretation is that the ark was designed to “run before the wind”, i.e. to drift downwind, without any propulsion system or steering device.

**The ark had to have directional stability:
It used the wind to maintain heading relative to
the wind:**

**It had a raised bow (Short Forecastle) to swing
the bow downwind**

**It had a skeg or underwater stabilizing fin back
aft to provide directional stability.**

Drifting Distance Estimate:

The ark could have drifted a maximum of:

$$150 \times 24 \times 2 = 7200 \text{ miles}$$

or just under 1/3 of the way around the world.

Hydrostatic Stability

Hydrostatic stability is governed by the Breadth/Depth (B / D) ratio.

Since the ark had a B/D ratio close to that of a typical cargo ship it would tend to have similar stability characteristics as well.

$$\mathbf{B/D = 5 / 3 = 1.667.}$$

Seakeeping

The ark's performance in waves.

Seas were infinite fetch.

The ark had to avoid getting into a beam seas situation where the rolling motions would cause a survival and habitability problem

It was designed to drift downwind (following seas), having a raised forecastle to swing the bow around and was fitted with a skeg aft to provide directional stability.

Directional Stability

The ability of a ship to run in a straight line, or maintain course without excessive rudder action..

Ark was fitted with a skeg aft to provide directional stability

Conclusions

General:

The ark was designed as a seaworthy oceangoing vessel capable of surviving the environmental conditions of the great flood, while accommodating the human crew and the large number of animals.

The ark's principal dimensions were very close to those of a typical cargo ship of the 1950's.

The ark was built around 2500 B. C. Therefore the ark's design and construction was about 4500 years ahead of its time.

Conclusions on Ark's Design:

The wood construction does not appear to be a problem from a structural strength standpoint, .

Ark was capable of carrying about 950 horse-size animals, or a combination of larger animals and thousands of smaller animals along with adequate food and water, assuming the animals were in a dormant hibernating-like state for most of the voyage.

The ark was designed to sail, navigate and negotiate the waves during the flood without any human intervention.

The ark was designed to “run before the wind”, i.e. to drift downwind, avoiding beam seas, negotiating the extreme waves without a rudder, steadying sails or sea anchors.

The ark's hull had a streamlined underwater shape with a definite bow and stern. It had a raised bow or short forecastle to swing the bow downwind and it had a skeg back aft to provide directional stability.

The one cubit “window” was a ventilation system consisting of shutters mounted on low deckhouses on the main deck.

The “covering” of the ark was the system of ventilator/deckhouses on the main deck. The roofs of the deckhouses were removed to revive the animals just prior to disembarkation.

Recommendations

Future proposed design verification study:

- **More detailed structural analysis and design work**
- **Weight and C.G. estimation**
- **Preparation of lines drawings**
- **Computation of hydrostatic characteristics including transverse stability.**
- **Directional stability analysis**
- **Seakeeping analysis**
- **Model construction**
- **Model testing in wind and waves to determine seakeeping characteristics**

Possible source of funding: NSF faith – based grant.

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